

AX88179B Product Introduction

Revision 1.02
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Revision History

Revision	Date	Description
1.00	2023/10/16	Initial release
1.01	2024/01/17	Updated Selection Guide in Figure 3.
1.02	2024/05/13	1. Updated Selection Guide in Figure 3. 2. Added AX88179BQI information in Section 4.

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1. Introduction

The AX88179B USB 3.2 Gen1 to 10/100/1000M Gigabit Ethernet controller is a high performance and highly integrated ASIC which enables low cost, small form factor, and simple plug-and-play Gigabit Ethernet network connection capability for desktops, notebook PC's, Ultrabook's, docking stations, game consoles, digital-home appliances, and any embedded system using a standard USB port.

The AX88179B features a USB interface to communicate with a USB Host Controller and is Compliant with USB specification V3.2 Gen1, V2.0, and V1.1. It implements a 10/100/1000Mbps Ethernet LAN function based on IEEE802.3, IEEE802.3u, and IEEE802.3ab standards with embedded SRAMs for packet buffering. And, it also integrates an on-chip 10/100/1000Mbps IEEE-compliant Ethernet PHY to simplify system design.

This document provides an overview of AX88179B USB 3.2 Gen1 to 10/100/1000M Gigabit Ethernet controller product.

2. Block Diagram

The following is AX88179B block diagram,

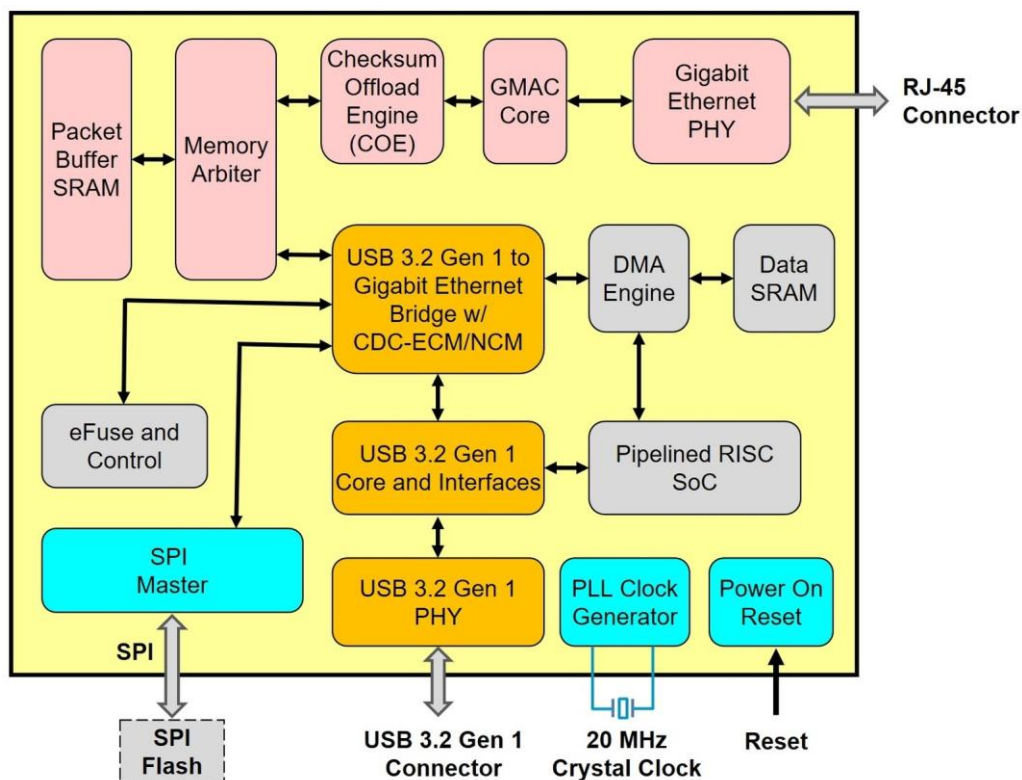


Figure 1. AX88179B Block Diagram

2-1. Typical System Block Diagram

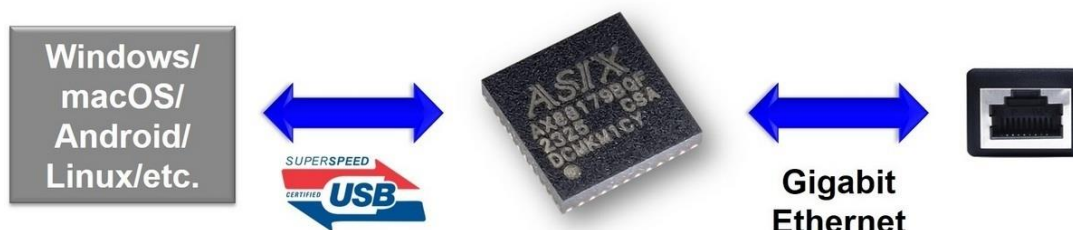


Figure 2. AX88179B Typical Applications Diagram

3. Selection Guide

The following is the selection guide of ASIX Electronics USB Ethernet ICs for different requirement applications. Please visit ASIX Electronics USB Ethernet ICs web pages and contact ASIX's Sales (sales@asix.com.tw) for details.

Part No.	USB Speed	Ethernet MAC/ PHY (bps)	CDC-NCM Driverless	iPXE	PTP	MAC Interface	Temperature Range (°C)	Package
AX88279	USB 3.2	2.5G/1G/100M	V	V	V	-	0 ~ +70	QFN-60
AX88179B	USB 3.2	1G/100M/10M	V	V	V	-	0 ~ +70 -40 ~ +85	QFN-40
AX88179A	USB 3.2	1G/100M/10M	V	V	-	-	0 ~ +70	QFN-40
AX88772E	USB 2.0	100M/10M	V	V	V	-	0 ~ +70	QFN-32
AX88772D	USB 2.0	100M/10M	V	V	-	-	0 ~ +70	QFN-32
AX88772C	USB 2.0	100M/10M	-	-	-	RMII/ Rev-RMII	0 ~ +70/ -40 ~ +85	LQFP-64
AX88772B	USB 2.0	100M/10M	-	-	-	RMII/ Rev-RMII	0 ~ +70/ -40 ~ +85	LQFP-64
AX88772A	USB 2.0	100M/10M	-	-	-	-	0 ~ +70	LQFP-64

Figure 3. Selection Guide

4. Ordering Information

The following are the ordering information of AX88179B silicon and AX88179B demo board. Please contact ASIX's Sales (sales@asix.com.tw) for more details.

Part Number	Description
AX88179BQF	40-pin QFN lead Free package, Commercial temperature range: 0 to 70°C.
AX88179BQI	40-pin QFN lead Free package, Industrial temperature range: -40 to +85°C

AX88179B Demo Board	Description
AX88179B Demo Board	This is a USB dongle for AX88179B USB 3.2 to Gigabit Ethernet application

Figure 4. Ordering Information

5. Target Applications

The following are some PC/Internet and consumer electronics target applications for your reference.

PC/Internet



Consumer Electronics



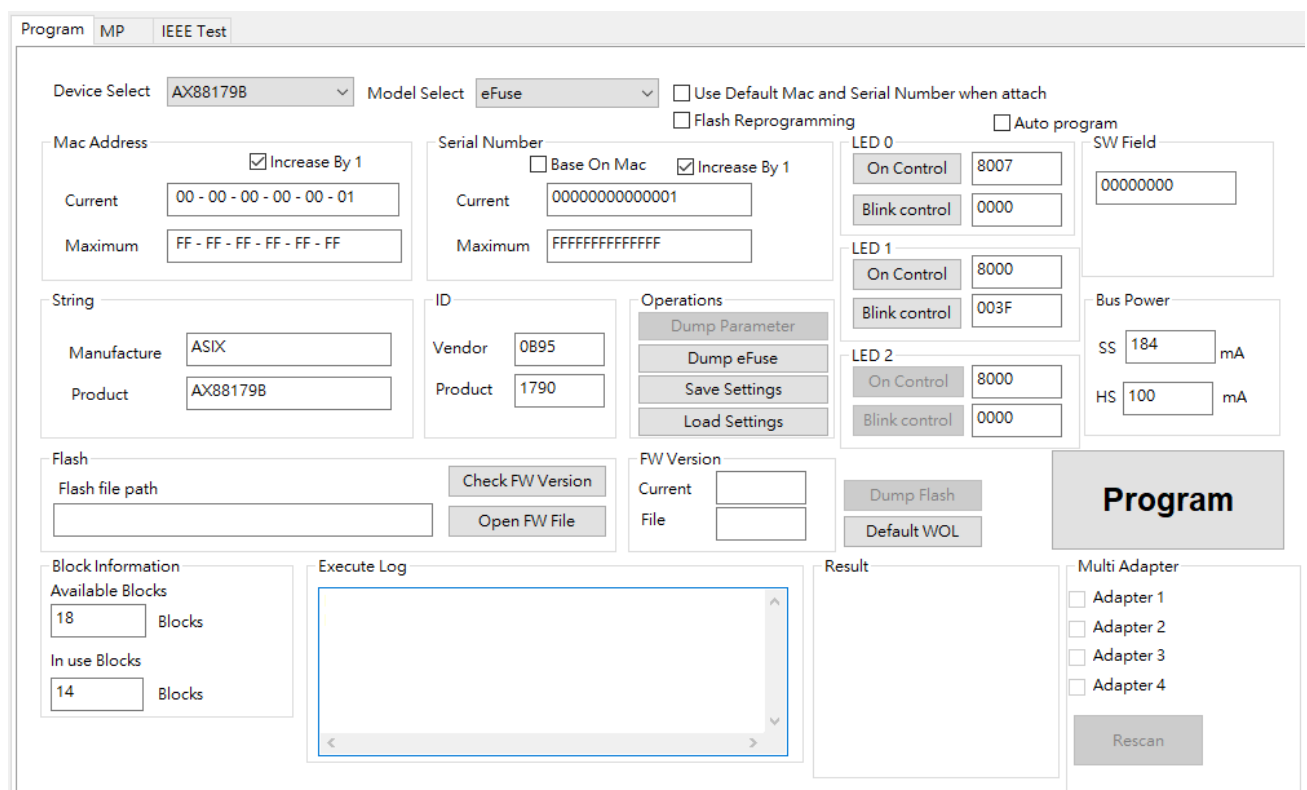
Figure 5. Target Applications

6. Mass Production Solutions

To support the mass production for those products using AX88179B chip, ASIX provides the Windows Mass Production Tool solution for AX88179B customers. This chapter provides a brief introduction for this solution. Please refer to “**AX88179B Mass Production User Guide**” for details.

6-1. Windows Mass Production Tool

ASIX Electronics provides a Windows Mass Production tool for users to easily program the Serial Flash/eFuse of AX88179B on a typical Windows PC. This AX88179B Windows Mass Production Tool supports to customize the MAC address, Serial Number, Vendor ID and Product ID, etc. for AX88179B based application systems in mass production.



The screenshot displays the 'Program' page of the Windows Mass Production Tool. The interface is organized into several functional areas:

- Top Navigation:** Tabs for 'Program', 'MP', and 'IEEE Test' are visible, with 'Program' being the active tab.
- Device and Model Selection:** 'Device Select' is set to 'AX88179B' and 'Model Select' is set to 'eFuse'. Checkboxes for 'Use Default Mac and Serial Number when attach' and 'Flash Reprogramming' are present.
- Mac Address Configuration:** Includes a checkbox for 'Increase By 1', a 'Current' field with the value '00 - 00 - 00 - 00 - 00 - 01', and a 'Maximum' field with the value 'FF - FF - FF - FF - FF - FF'.
- Serial Number Configuration:** Includes a checkbox for 'Base On Mac', a checkbox for 'Increase By 1', a 'Current' field with the value '000000000000001', and a 'Maximum' field with the value 'FFFFFFFFFFFFFF'.
- String Configuration:** Fields for 'Manufacture' (ASIX) and 'Product' (AX88179B).
- ID Configuration:** Fields for 'Vendor' (0B95) and 'Product' (1790).
- Operations Panel:** Buttons for 'Dump Parameter', 'Dump eFuse', 'Save Settings', and 'Load Settings'.
- LED Control:** Sections for LED 0, LED 1, and LED 2, each with 'On Control' and 'Blink control' fields.
- SW Field:** A field for 'SW Field' with the value '00000000'.
- Bus Power:** Fields for 'SS' (184 mA) and 'HS' (100 mA).
- Flash Configuration:** A 'Flash file path' field, 'Check FW Version' and 'Open FW File' buttons, and a 'FW Version' section with 'Current' and 'File' fields.
- Block Information:** 'Available Blocks' (18) and 'In use Blocks' (14).
- Execute Log:** A large text area for logging execution details.
- Result:** A section for displaying the results of the operation.
- Multi Adapter:** Checkboxes for 'Adapter 1', 'Adapter 2', 'Adapter 3', and 'Adapter 4', along with a 'Rescan' button.
- Program Button:** A large 'Program' button to initiate the mass production process.

Figure 6. Program Page for Windows Mass Production Tool

This tool also supports to send/receive packets in different Ethernet speed modes, and program the serial Flash/eFuse. This tool can be run on a Windows PC, which installs the AX88179B Windows driver. This tool also needs a separate server PC to run the test server tool (TestServer.exe). The test server tool on server PC can receive packets from the “device under test” product, and then reply back.

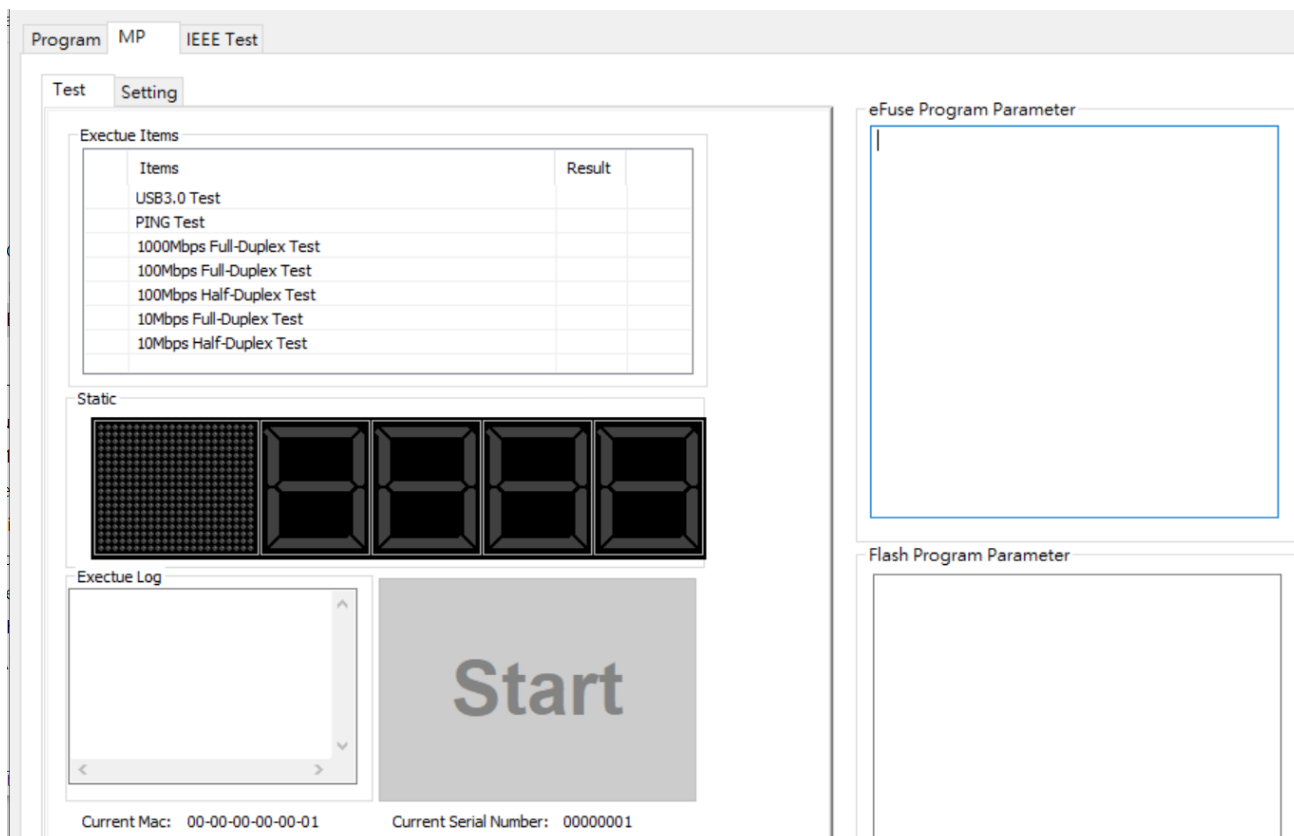


Figure 7. MP Page for Windows Mass Production Tool

7. AX88179B Demo Board

ASIX Electronics provides AX88179B demo board for users to evaluate the basic functions of AX88179B on different target applications. If you need to purchase the AX88179B demo board, please contact ASIX's Sales (sales@asix.com.tw) for more details.



Figure 8. AX88179B Demo Board



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